

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

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FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.7: VOLUME AND CAPACITY

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent the last 8 lessons investigating why the tea farmers' containers in Kericho — a cylindrical drum, a conical sisal cover, and a frustum-shaped metal bucket — look nearly the same size but hold such dramatically different amounts of water. Now it is your turn to bring all of your learning together into one well-reasoned, mathematically precise explanation.

Read the driving question carefully before you begin:

"Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?"

Your final explanation must do three things:

1. Use correct mathematical vocabulary and formulas to explain the volume relationships between a cylinder, a cone, a pyramid, and a frustum.
2. Connect every formula back to the Kericho tea farmers' phenomenon — do not explain in the abstract only.
3. Show a complete, worked design calculation for a frustum-shaped container that meets a specific capacity target.

Work through each section below in order. Write in full sentences where prompted. Show all working where calculations are required. Use units consistently (cm, cm³, litres) throughout.

Section 1: The Cylinder–Cone Relationship — Why the Drum Holds Three Times More

Explain, in your own words and using the correct formulas, why a cylinder and a cone with the same base radius and the same height do NOT hold the same amount. Then apply this to the Kericho farmers' containers.

- (a) Write the formula for the volume of a cylinder and the formula for the volume of a cone. Define every variable.
- (b) Show algebraically why $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$ when the radius r and height h are equal for both shapes.
- (c) The cylindrical drum used by the farmers has a radius of 30 cm and a height of 60 cm. Calculate its volume in cm³ and convert your answer to litres (1 litre = 1 000 cm³).
- (d) Using only your answer from (c), state — without recalculating from scratch — what volume of water the conical sisal cover holds. Explain your reasoning in one sentence.

The volume of a cylinder is $V = \pi r^2 h$, where r is the base radius and h is the perpendicular height. The volume of a cone with the same base and height is $V = \frac{1}{3} \pi r^2 h$. Dividing one by the other: $(\pi r^2 h) \div (\frac{1}{3} \pi r^2 h) = 3$, which proves that the cylinder always holds exactly three times as much as the cone, regardless of the actual size, as long as r and h are identical.

For the Kericho farmers' cylindrical drum: $V = \pi \times 30^2 \times 60 = \pi \times 900 \times 60 = 54\,000\pi \approx 169\,646 \text{ cm}^3 \approx 169.6 \text{ litres}$.

The conical sisal cover therefore holds $169\,646 \div 3 \approx 56\,549 \text{ cm}^3 \approx 56.5 \text{ litres}$, because the cone's volume is always one-third of a cylinder sharing the same radius and height. This is why the cylindrical drum looks only slightly taller and wider to the eye, yet the farmers can fill it with three full loads of water that would exactly fill the cone — the difference is entirely due to the shape of the interior space, not the exterior appearance.

Section 2: Extending the Pattern — Pyramids and Their Relationship to Prisms

The one-third relationship is not unique to cones and cylinders. It also holds between pyramids and prisms. Demonstrate this and connect it back to the farmers' understanding of

The volume of any pyramid is $V = \frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} B h$, where B is the area of the base (any shape) and h is the perpendicular height from base to apex.

Square prism (crate): $V = 40 \times 40 \times 50 = 80\,000 \text{ cm}^3$.

Square pyramid (lid): $V = \frac{1}{3} \times 40 \times 40 \times 50 = \frac{1}{3} \times 80\,000 \approx 26\,667 \text{ cm}^3$.

capacity. (a) Write the general formula for the volume of any pyramid. State what each variable represents. (b) A square-based storage crate used at the Kericho tea collection centre measures 40 cm × 40 cm at the base and is 50 cm tall (a square prism). A decorative pyramid-shaped lid has the same square base and the same height. Calculate the volume of both the crate and the pyramid-lid. Show full working. (c) What fraction of the crate's volume does the pyramid-lid represent? Does this surprise you? Explain why or why not, linking your answer to what you discovered about cones and cylinders in Section 1. (d) In one or two sentences, explain why understanding this pattern helps a farmer or a container designer predict capacity without always measuring by filling with water.	The pyramid-lid holds exactly $\frac{1}{3}$ of the crate's volume — the same one-third fraction seen between the cone and cylinder. This is not a coincidence. It is a universal geometric truth: any pointed solid (cone or pyramid) holds exactly one-third the volume of its matching flat-topped solid (cylinder or prism) when base and height are equal. A farmer or designer who knows this rule can immediately estimate capacity from dimensions alone, saving time and avoiding the wasteful process of trial-and-error filling — which matters greatly when designing containers to carry exact weights of tea leaves to the weighing station.
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Section 3: Understanding the Frustum — Why the Metal Bucket Sits Between the Cone and the Cylinder

The farmers' metal bucket is a frustum — a cone with its top cut off. This is why it holds more than the cone but less than the cylinder. Use the frustum volume formula to explain and calculate. (a) Explain in your own words what a frustum is and how it is formed from a cone. Use a	A frustum is what remains when you slice a cone parallel to its base and remove the smaller cone on top. It looks like a cone with its pointed tip cut off — exactly the shape of the farmers' metal bucket, which is wide at the bottom and narrower at the top opening. The frustum formula $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ is derived by calculating the volume of the full large cone (height H, radius R) and subtracting the volume of the small cone that was removed (height $H - h$, radius r). After applying similar-triangle ratios and simplifying, the result is the formula above. For the farmers' bucket: $V = \frac{1}{3} \times \pi \times 40 \times (25^2 + 25 \times 15 + 15^2)$ $= \frac{1}{3} \times \pi \times 40 \times (625 + 375 + 225)$
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labelled sketch or a clear written description to show the large radius R, the small radius r, and the height h of the frustum. (b) State the formula for the volume of a frustum: $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. Explain where this formula comes from — hint: think about subtracting a smaller cone from a larger cone. (c) The farmers' metal bucket (frustum) has a large base radius $R = 25$ cm, a small top radius $r = 15$ cm, and a height $h = 40$ cm. Calculate its volume in cm^3 and convert to litres. Show full working. (d) Compare your answer to the cone volume from Section 1 (same height of 40 cm, radius 25 cm) and explain in one sentence why the frustum holds more than that cone.	$= \frac{1}{3} \times \pi \times 40 \times 1\,225$ $= \frac{1}{3} \times 49\,000\pi$ $\approx \frac{1}{3} \times 153\,938$ $\approx 51\,313\text{ cm}^3$ $\approx 51.3\text{ litres.}$ A cone with the same height (40 cm) and base radius (25 cm) would hold only $V = \frac{1}{3} \times \pi \times 25^2 \times 40 = \frac{1}{3} \times \pi \times 25\,000 \approx 26\,180\text{ cm}^3 \approx 26.2\text{ litres}$. The frustum holds nearly double that of the equivalent cone because removing the tip of a cone and widening that section replaces empty pointed space with actual usable volume — the farmers' bucket is essentially a cone that has had its least-useful, narrowest portion replaced with a wider, more capacious top opening.
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Section 4: Designing a Frustum Container for the Farmers — A Full Mathematical Design

A tea cooperative in Kericho wants to commission a new frustum-shaped aluminium container that holds exactly 80 litres of water ($80\,000\text{ cm}^3$). The manufacturer sets two fixed constraints: the height must be $h = 50$ cm, and the ratio of the top radius to the bottom radius must be $r = 0.6R$ (the top opening is 60% of the base width). Your task is to find the value of R (the base radius) that satisfies these constraints, then verify your design.	Step (a) — Substituting $r = 0.6R$ and $h = 50$: $R^2 + Rr + r^2 = R^2 + R(0.6R) + (0.6R)^2 = R^2 + 0.6R^2 + 0.36R^2 = 1.96R^2.$ $\text{So } V = \frac{1}{3} \times \pi \times 50 \times 1.96R^2 = (\frac{1}{3} \times 50 \times 1.96\pi)R^2 = (98\pi/3)R^2.$ Step (b) — Setting equal to 80 000: $(98\pi/3)R^2 = 80\,000$ $R^2 = (80\,000 \times 3) / (98\pi)$ $R^2 = 240\,000 / (307.88)$ $R^2 \approx 779.5$ $R \approx \sqrt{779.5} \approx 27.9\text{ cm.}$ Step (c) — Design values: $R \approx 27.9$ cm; $r = 0.6 \times 27.9 \approx 16.7$ cm. Verification: $V = \frac{1}{3} \times \pi \times 50 \times (27.9^2 + 27.9 \times 16.7 + 16.7^2)$ $= \frac{1}{3} \times \pi \times 50 \times (778.4 + 465.9 + 278.9)$
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(a) Substitute $r = 0.6R$ and $h = 50$ into the frustum formula $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ and simplify the expression inside the brackets in terms of R only. (b) Set your simplified expression equal to $80\,000\text{ cm}^3$ and solve for R. Show every algebraic step. Round to 1 decimal place. (c) State the values of R and r for your designed container. Then verify your answer by substituting both values back into the original frustum formula and confirming you get approximately $80\,000\text{ cm}^3$. (d) Convert: how many 20-litre jerricans of water would this container fill when full? Is this a practical size for a tea farmer to carry? Explain your reasoning using context from the Kericho phenomenon.	$= \frac{1}{3} \times \pi \times 50 \times 1\,523.2$ $= \frac{1}{3} \times 239\,496$ $\approx 79\,832\text{ cm}^3 \approx 80\text{ litres. } \checkmark \text{ (Small rounding difference is acceptable.)}$ Step (d) — $80\text{ litres} \div 20\text{ litres} = 4$ jerricans. This is a practical but heavy load — a full container of water at 80 kg would not be carried by one person. However, the farmers do not fill it to the brim with water; they use it to carry harvested tea leaves, which are far lighter than water for the same volume. Knowing the water capacity allows the cooperative to calculate the maximum leaf load by accounting for the lower density of tea leaves, making the mathematics directly useful on the Kericho farm.
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Section 5: Answering the Driving Question — Your Complete Explanation

You have now worked through all the mathematics. In this final section, write a coherent, well-structured explanation that directly and fully answers the driving question. Your response should be approximately 200–300 words. Your explanation must: – State clearly why the three containers look similar but hold different volumes, using the correct mathematical relationships. – Reference the specific	The three containers in Kericho — the cylindrical drum, the conical sisal cover, and the frustum-shaped metal bucket — appear similar in size because they share nearly the same height and the same opening diameter. However, shape, not size, determines how much space exists inside a container, and the three shapes distribute their interior space in fundamentally different ways. The cylinder fills every cross-section from base to top with the same circular area, giving it a volume of $V = \pi r^2 h$. The cone, by contrast, tapers continuously to a point, so most of its height contributes very little volume — mathematically, this tapering reduces its capacity to exactly one-third of the cylinder's, giving $V = \frac{1}{3}\pi r^2 h$. This is why the farmers' cylindrical drum holds precisely three times as much water as the conical sisal cover, even though both containers stand at the same height and have the same base opening. The eye is deceived because it reads the outline, not the interior. The frustum-shaped metal bucket sits between the two extremes: it begins as a cone but has its narrow top removed and replaced with a wider opening. By keeping more of the wider, high-volume lower section and discarding the near-empty pointed tip, the frustum reclaims much of the volume the cone loses, giving it $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. This is why the bucket holds noticeably more than the cone but still less than the full cylinder.
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formulas for the cylinder, cone, and frustum. – Explain the role of shape — not just size — in determining capacity. – Connect your mathematical reasoning directly to the Kericho tea farmers and their everyday work. – End with a statement about how mathematics empowers the farmers or cooperative to make better decisions about container design. Do NOT simply repeat your calculations. This section is about communicating understanding clearly and convincingly.	Most powerfully, once the farmers or the cooperative understand these formulas, they are no longer guessing. They can calculate the exact dimensions needed for a container of any required capacity — as demonstrated in the design task for the 80-litre aluminium frustum. Mathematics transforms a puzzling observation on a Kericho farm into a precise engineering tool, saving materials, reducing waste, and ensuring every container does exactly the job it was built to do.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Mathematical Accuracy — Formulas and Calculations	All formulas (cylinder, cone, pyramid, frustum) are stated correctly with variables defined. All calculations are accurate, fully shown step-by-step, and consistently use correct units (cm, cm ³ , litres). Rounding is appropriate and stated.	Formulas are mostly correct and calculations are shown with only minor arithmetic errors that do not affect overall reasoning. Units are mostly consistent with one or two omissions.	Some formulas are missing or incorrect. Calculations contain significant errors or missing steps. Units are inconsistently used or omitted, making it difficult to follow the working.
Conceptual Understanding — Shape, Volume and the One-Third Relationship	The student clearly and accurately explains why a cone holds one-third the volume of a cylinder with the same base and height, extends this reasoning correctly to pyramids and prisms, and articulates why the frustum sits between the cone and cylinder. Explanation goes beyond restating formulas to show genuine understanding of why the relationship exists.	The student explains the one-third relationship and applies it correctly in most cases. The explanation of the frustum's position between cone and cylinder is present but may lack depth or full algebraic justification.	The student identifies that the containers hold different amounts but cannot clearly explain the mathematical reason. The one-third relationship is stated without justification, or is confused between shapes.
Design Task — Algebraic	The student correctly substitutes $r = 0.6R$	The student sets up the substitution	The student attempts the substitution but

Reasoning and Problem Solving	into the frustum formula, simplifies fully and accurately, solves for R showing all algebraic steps, and verifies the answer by back-substitution. The solution is clearly communicated and the result is interpreted in context (jerricans, practicality for farmers).	correctly and reaches a value for R with only minor errors. Verification is attempted. Contextual interpretation is present but may be brief or partially correct.	makes significant algebraic errors that lead to an incorrect value of R. Verification is missing or incorrect. Little or no contextual interpretation is provided.
Connections to the Kericho Phenomenon and Real-World Context	Every section explicitly and specifically connects mathematical results back to the Kericho tea farmers' containers and their work. The student uses the farmers' actual container dimensions and explains how the mathematics is useful in a farming or cooperative context. Kenyan context feels authentic, not decorative.	Most sections reference the Kericho phenomenon. The connections are present and relevant but may be formulaic or limited to one or two sentences per section rather than deeply integrated throughout.	References to the Kericho context are rare, vague, or only appear in the final section. The explanation could have been written about any containers anywhere, with the Kenyan context added superficially.
Communication — Clarity, Structure and Mathematical Language	The final explanation (Section 5) is well-structured, clearly written, and fully answers the driving question in 200–300 words. Mathematical vocabulary (frustum, radius, capacity, volume, cross-section, taper) is used correctly and consistently throughout all five sections. The writing is confident and precise.	The final explanation addresses the driving question adequately and uses most mathematical vocabulary correctly. Minor gaps in structure or vocabulary do not significantly impede understanding.	The final explanation is incomplete, unclear, or does not directly address the driving question. Mathematical vocabulary is limited, misused, or absent, making it difficult to assess the student's understanding from their writing alone.